

The Evolution of the Adaptive Horizon Sniper Engine: From Naive Prediction to Regime-Aware Quantitative Research

Abstract

Financial machine learning systems frequently report strong historical performance that fails to survive realistic evaluation. Common causes include overfitting, data leakage, probability miscalibration, and unrealistic execution assumptions. Consequently, backtest returns alone often provide little evidence of genuine predictive capability.

This paper documents the development of the Adaptive Horizon Sniper Engine, a quantitative research framework for short-term cryptocurrency prediction. The project evolved through five major versions, each designed to address weaknesses identified in previous iterations. The progression included walk-forward validation, probability calibration, execution-cost modeling, cross-asset testing, regime analysis, and ultimately a regime-aware expert ensemble architecture.

The results demonstrate that increasingly rigorous validation procedures substantially reduce apparent performance while improving confidence in the legitimacy of observed signals. They also suggest that predictive behavior is strongly regime-dependent and that specialized models may offer greater stability than monolithic forecasting systems. The primary contribution of this work is not a trading strategy, but a documented research process showing how successive methodological improvements transformed an initially naive predictive model into a more robust quantitative research framework.

1. Introduction

Financial markets are among the most difficult environments for machine learning. Market behavior is noisy, adaptive, and non-stationary. Relationships that appear predictive during one period often weaken or disappear as conditions change.

Recent advances in machine learning have produced a growing number of financial forecasting systems. Many report impressive historical performance, yet relatively few survive realistic validation. In practice, strong backtests are often the result of data leakage, overfitting, unrealistic execution assumptions, or inadequate evaluation procedures rather than genuine predictive power.

This project began with a simple question: can machine learning identify meaningful short-term directional opportunities in cryptocurrency markets? Rather than pursuing maximum historical return, the objective was to determine which forms of predictive performance remained valid under increasingly strict testing conditions.

Over time, the project evolved through multiple architectural generations. Early versions focused on prediction itself. Later versions focused on calibration, execution realism, robustness, and generalization. The final framework adopted a regime-aware architecture in which specialized models operate under different market conditions rather than relying on a single forecasting engine.

The development process revealed a recurring pattern. Many improvements that appeared beneficial under loose evaluation standards disappeared when tested more rigorously. Conversely, several techniques that reduced apparent performance increased confidence in the validity of the remaining results. This observation became the central theme of the research.

The purpose of this paper is to document that evolution. Rather than presenting a finished trading system, it examines the sequence of assumptions, failures, modifications, and discoveries that shaped the development of the Adaptive Horizon Sniper Engine.

2. Research Questions

The project was guided by three primary research questions.

First, can machine learning models identify short-term directional structure in cryptocurrency markets that remains detectable on unseen data?

Second, can any observed predictive edge survive realistic evaluation procedures, including walk-forward testing, probability calibration, transaction costs, slippage, and robustness analysis?

Third, does predictive performance depend on market regime, and if so, can specialized models operating under different market conditions outperform a single generalized forecasting system?

These questions remained consistent throughout the project and motivated every major architectural revision from Version 1 through Version 5.

3. Framework Evolution

The Adaptive Horizon Sniper Engine evolved through five major versions. Each version was developed in response to a specific weakness identified in previous experiments. Rather than pursuing performance through increasing complexity alone, the project followed an iterative research process in which assumptions were continuously tested and revised. As validation standards became more demanding, the framework gradually shifted from a simple predictive model toward a regime-aware quantitative research system.

3.1 Version 1: The Primitive Baseline

Version 1 was designed to establish a complete machine learning pipeline capable of ingesting historical cryptocurrency data, generating features, training predictive models, and

producing forecasts. The framework relied on OHLCV market data and a Random Forest model to predict future price behavior.

Initial experiments produced strong training performance but little practical value. The system suffered from methodological weaknesses including feature leakage, inadequate validation procedures, and unrealistic execution assumptions. While the model appeared successful on historical data, it failed to demonstrate reliable out-of-sample performance.

The primary lesson of Version 1 was that predictive accuracy alone provides limited evidence of genuine forecasting ability. This motivated a transition toward more rigorous validation methods.

3.2 Version 2: Walk-Forward Learning

Version 2 addressed the problem of non-stationarity. Financial markets evolve continuously, making static train-test splits an unreliable measure of future performance.

To account for this, the framework adopted a rolling walk-forward architecture built around a LightGBM classifier. The model was retrained repeatedly using recent observations before generating future predictions, creating a validation environment that more closely resembled real deployment.

This change improved stability and reduced dependence on outdated historical relationships. However, it exposed a new weakness. Although the model often ranked opportunities effectively, its confidence estimates were unreliable. Highly confident predictions frequently failed to achieve the expected success rates.

The distinction between prediction quality and probability calibration became the central challenge for the next stage of development.

3.3 Version 3: Calibration and Execution Realism

Version 3 focused on improving the realism of the evaluation process. Probability calibration was introduced through Platt Scaling, allowing model outputs to be interpreted more consistently as confidence estimates. Transaction costs and ATR-adjusted slippage were also incorporated into the testing framework.

These additions significantly reduced apparent performance. Trade frequency declined, many marginal signals disappeared, and reported returns became less impressive than those observed in earlier versions. At the same time, confidence in the remaining results increased substantially.

Version 3 demonstrated an important principle that would shape the remainder of the project: removing unrealistic assumptions often lowers performance metrics while improving the credibility of the conclusions. The objective increasingly shifted away from maximizing returns and toward identifying effects that could survive realistic testing conditions.

3.4 Version 4: Generalization and Regime Analysis

After improving validation quality, the next challenge was determining whether the framework's behavior generalized beyond a single asset. Previous experiments left open the possibility that observed performance was tied to the specific characteristics of a limited set of cryptocurrencies.

Version 4 expanded testing across multiple assets, including Bitcoin, Ethereum, Solana, Binance Coin, and XRP. A common parameter configuration was evaluated across the entire universe without asset-specific optimization. In parallel, market-regime profiling was introduced to analyze performance under different volatility and trend conditions.

Cross-sectional testing provided evidence that some predictive behaviors generalized across assets. However, regime analysis revealed a more significant finding. Performance was not evenly distributed across market environments. The framework consistently performed better during bearish and high-volatility conditions than during bullish or low-volatility periods.

This observation challenged a core assumption underlying earlier versions. Rather than functioning as a universal forecasting system, the model appeared to specialize in specific forms of market behavior. The results suggested that different market regimes may require different predictive mechanisms.

3.5 Version 5: Regime-Aware Expert Ensembles

Version 5 represents the culmination of the project's development. Instead of attempting to improve a single forecasting model, the architecture was redesigned around the idea that market regimes should be treated as distinct prediction problems.

A rolling Hidden Markov Model was introduced as a regime dispatcher responsible for identifying latent market states. Observations were then routed to specialized expert models optimized for different environments. These included momentum, mean-reversion, volatility-expansion, and risk-mitigation experts.

The framework also incorporated purged walk-forward validation, Bayesian hyperparameter optimization, feature-selection mechanisms, probability calibration layers, dynamic execution-cost modeling, and block-bootstrap robustness analysis. These components were integrated into a unified research pipeline designed to reduce data leakage, overfitting, and backtest inflation.

The transition from a monolithic model to a regime-aware ensemble architecture represents the most significant conceptual change in the project's history. Earlier versions attempted to adapt a single model to all market conditions. Version 5 instead assumes that markets exhibit multiple statistical regimes and assigns specialized decision-making systems to each one.

Across all five versions, the project's most important finding was not the discovery of a specific trading rule. It was the recognition that predictive performance in financial markets is highly dependent on validation quality, execution realism, and regime awareness. The evolution of the framework reflects a gradual movement away from maximizing historical performance and toward building a more reliable process for quantitative research.

4. Results

The objective of the experimental evaluation was not to identify the highest-performing model, but to determine how predictive performance changed as increasingly rigorous validation procedures were introduced. Each version of the Adaptive Horizon Sniper Engine was therefore assessed not only by its reported returns, but also by its robustness, stability, and ability to generalize across changing market conditions.

The progression from Version 1 through Version 5 revealed a consistent pattern. Early versions frequently produced attractive performance metrics under simplified assumptions, but these results deteriorated when more realistic testing procedures were applied. Transaction costs, slippage, calibration requirements, and stricter validation protocols reduced apparent profitability while improving confidence in the validity of the remaining signals.

Version 2 demonstrated that walk-forward learning could maintain predictive performance more effectively than static training procedures. The framework consistently outperformed random directional prediction across multiple evaluation windows, suggesting that short-term market structure contained information exploitable by machine learning models. However, the reliability of confidence estimates remained limited.

Version 3 produced lower raw returns than Version 2 following the introduction of probability calibration and execution-cost modeling. Despite this reduction, risk-adjusted performance improved. The framework generated fewer trades, exhibited greater selectivity, and demonstrated more consistent behavior across testing periods. These results suggested that a significant portion of the apparent edge observed in earlier versions originated from optimistic assumptions rather than genuine predictive power.

Version 4 extended the evaluation across multiple cryptocurrencies and market environments. Cross-sectional testing indicated that predictive performance generalized beyond a single asset, providing evidence that the framework was identifying broader market behaviors rather than memorizing the characteristics of one dataset. At the same time, regime analysis revealed substantial variation in performance across market conditions. Predictive accuracy was strongest during bearish and high-volatility environments and weaker during bullish or range-bound periods.

The most significant result emerged from this regime analysis. Rather than behaving as a universal forecasting system, the framework appeared to specialize in particular market states. This observation directly motivated the architectural redesign introduced in Version 5.

Early testing of the Version 5 architecture suggests that regime-aware routing improves performance stability relative to monolithic forecasting approaches. By assigning observations to specialized expert models, the framework is able to adapt its decision-making process to changing market environments. While research remains ongoing, preliminary results indicate improved consistency across regimes and reduced sensitivity to abrupt market transitions.

Taken together, the results support three primary conclusions. First, machine learning models can identify statistically meaningful short-term patterns within cryptocurrency markets. Second, much of the apparent performance observed under simplified testing conditions disappears when realistic constraints are introduced. Third, regime awareness appears to play a critical role in achieving stable predictive behavior across changing market environments.

Versio n	Major Innovation	Primary Finding
V1	Baseline Random Forest	High in-sample performance did not translate to reliable out-of-sample behavior
V2	Walk-Forward LightGBM	Continuous retraining improved stability under non-stationary conditions
V3	Calibration & Execution Modeling	Realistic assumptions reduced returns but increased result credibility
V4	Cross-Asset Validation	Predictive behavior generalized across assets but exhibited regime dependence
V5	Regime-Aware Expert Ensemble	Specialized models improved stability across changing market states

Asset	Return	Precision	Alpha vs B&H
BTC	+72.65%	63.3%	+79.37%
ETH	+102.18%	63.9%	+139.65%

SOL	+36.84%	57.8%	+89.08%
XRP	+150.03%	57.8%	+57.54%
ADA	+82.37%	60.0%	+143.15%
AVAX	+106.74%	64.6%	+182.92%
LINK	+1.73%	56.1%	+39.10%
DOT	+77.02%	55.6%	+157.89%

5. Discussion

The development of the Adaptive Horizon Sniper Engine produced several insights that extend beyond the specific framework presented in this paper.

The first concerns validation. Throughout the project, the largest performance reductions did not result from changes in model architecture. Instead, they resulted from improvements in evaluation methodology. Walk-forward testing, probability calibration, transaction-cost modeling, and robustness analysis consistently reduced reported performance while increasing confidence in the legitimacy of the results. This suggests that validation quality may be more important than model complexity in financial machine learning research.

A second observation involves feature engineering. Early experimentation explored increasingly large feature sets under the assumption that additional information would improve predictive performance. In practice, the opposite frequently occurred. Simpler models built upon a small number of carefully selected features consistently demonstrated greater robustness than models relying on large collections of indicators. This finding aligns with the broader challenge of overfitting in high-dimensional financial datasets.

The project also highlighted the importance of probability calibration. Accurate ranking of opportunities proved insufficient when confidence estimates were unreliable. Calibration transformed model outputs from simple scores into more meaningful measures of uncertainty, enabling improved decision-making and risk management. In many cases,

improvements in calibration contributed more to practical performance than improvements in raw predictive accuracy.

The most significant insight emerged from the regime analysis conducted during Version 4 and expanded in Version 5. Market behavior was not uniform across time. Different environments exhibited distinct statistical characteristics, and models that performed well in one regime frequently struggled in another. This observation challenged the assumption that a single predictive framework could operate optimally under all conditions.

The transition to a regime-aware expert architecture was therefore not simply a technical modification. It represented a change in how the prediction problem itself was defined. Rather than treating financial markets as a single forecasting task, the framework began treating them as a collection of related but distinct environments requiring specialized decision-making systems.

Viewed collectively, the evolution of the project suggests that successful quantitative research depends less on discovering a single powerful model and more on constructing a process capable of continuously identifying, testing, and correcting its own assumptions. The most valuable outcome of the project was not a particular algorithm or performance metric, but the development of a methodology for evaluating predictive systems under increasingly realistic conditions.

6. Limitations and Future Work

Despite the methodological improvements introduced throughout the project, several limitations remain. The framework was developed and evaluated using historical cryptocurrency data, and therefore remains vulnerable to structural changes that may occur in future market environments. Relationships identified during one period may weaken or disappear as market participants, liquidity conditions, and trading behavior evolve.

The current implementation also relies primarily on technical and market-derived features. While these variables capture aspects of price action, volatility, and liquidity, they do not incorporate broader sources of information such as macroeconomic conditions, derivatives positioning, order-book dynamics, or market sentiment. Future research could investigate whether these additional data sources improve regime identification and predictive performance.

Although significant effort was devoted to reducing backtest inflation through calibration, execution modeling, and robustness testing, historical simulation cannot fully replicate live deployment. Real-world trading introduces operational constraints and behavioral effects that are difficult to capture within a research environment. Consequently, future work should focus on forward testing and live evaluation under controlled conditions.

Version 5 represents the current stage of development rather than a final solution. Ongoing research is focused on improving regime detection, refining expert specialization, and further strengthening the framework's ability to adapt to changing market conditions.

7. Conclusion

This paper documented the evolution of the Adaptive Horizon Sniper Engine from a simple machine learning baseline into a regime-aware quantitative research framework. Across five major versions, the project progressed through a series of methodological improvements including walk-forward validation, probability calibration, execution-cost modeling, cross-asset testing, regime analysis, and expert-based model specialization.

The most important finding was not the discovery of a particular forecasting model, but the observation that increasingly rigorous evaluation procedures often produced conclusions very different from those suggested by naive backtesting. Many apparent sources of predictive performance weakened under stricter testing, while a smaller set of effects remained consistently observable.

The research also demonstrated the importance of regime awareness. Market behavior was found to vary substantially across different environments, limiting the effectiveness of monolithic forecasting approaches. This insight motivated the development of a regime-aware expert architecture designed to align specialized predictive models with specific market states.

Ultimately, the project evolved from an attempt to predict market direction into an investigation of how predictive systems should be evaluated. The resulting framework reflects a broader lesson in quantitative research: robust methodology, realistic assumptions, and continuous critical evaluation are often more valuable than raw performance metrics alone.